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Cloud Usage for Manufacturing: Challenges and Opportunities

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Abstract

Due to its potential to improve agility, efficiency, innovation, and global collaboration, cloud manufacturing is probably going to be an essential part of the manufacturing sector in the future. In a manufacturing environment that is becoming more and more competitive and changing quickly, businesses that strategically use cloud technologies could effectively broaden their capabilities. This research review demonstrates the possible development of cloud computing transformative effects on the manufacturing sector and highlights the advancements in cloud usage for manufacturing. The research examines various manufacturing uses for cloud computing, including cooperative production, data management, analytics, process optimization, and supply chain management. It also delves into some disadvantages of using the cloud in the manufacturing industry and estimates future developments in cloud-based manufacturing solutions. Finally, a simple use case is shown, using a typical commercial application.

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1. Introduction

Lately, complex issues in manufacturing have been resolved by using all forms of machine learning (ML) [1][2]. The support of computing and sensing technologies is what has made machine learning applications successful in the manufacturing industry. The field is already experiencing positive effects, as artificial intelligence and data-driven approaches advance more, together with the advancement of processing power [3]. The management of manufacturing data includes the storage, the process and the subsequent generation of results of this process. Before cloud computing emerged, this data has mainly been processed locally, in servers/computers based in the location of the manufacturing plant.

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The usage of Cloud Servers for manufacturing, to manage the data generated from all kinds of machinery in factories, has lately started to be examined by researchers and the industry, as an alternative to using local servers [4] [5]. The main advantages of Cloud Computing, for the applications in Manufacturing, include the flexibility of the system, the scalability, the current reliability, the security and finally, the collaboration capabilities. The cloud server operations can scale together with the manufacturing plant, providing a rapid decision system.

The limited computational capacity of local servers was no longer able to manage the tension between shorter time-to-market cycles, increased quality requirements, more complicated supply chains and the administration that was needed by the production staff. In addition to increased scalability and agility, cloud-based manufacturing operations also can provide process optimization and increased transparency across different manufacturing facilities.

The objectives of this study include examining the research history on cloud manufacturing, analyzing the opportunities and risks, looking at current commercial applications, and discussing potential future developments.

The following sections mention initially the key aspects of cloud manufacturing. After that, a background on the previous research efforts on the topic is stated. A use case in a commercial application is analyzed afterwards, and then the risks are stated in the fifth chapter. Finally, the conclusions are mentioned, along with some possible future research topics.

Nomenclature

ML	Machine Learning
IIoT	Industrial Internet of Things
IoT	Internet of Things
MaaS	Manufacturing as a Service
P2P	Peer to Peer
3D	Three Dimensional
CMfg	Cloud Manufacturing
QoS	Quality of Service
SM	Social Manufacturing
6G-IoV	6th Generation Internet of Verticals
SDCM	Software Defined Cloud Manufacturing
ETL	Extract, Transform, Load
AI	Artificial Intelligence
CNC	Computer numerical control

2. Key Aspects

The main key aspects of Cloud Computing [6], coupled with Manufacturing topics, are:

2.1. Supply chain management

Real-time visibility, collaboration, and optimization are made possible by cloud-based supply chain platforms. Inventory management, shipment tracking, demand forecasting, and improved customer and supplier communication are all features that manufacturers can improve [7].

2.2. Digital Twins

Digital twins, which are virtual replicas of real assets, procedures, or systems, can be created and managed with the help of cloud infrastructure. Digital twins can be used by manufacturers to test new designs, simulate and opti-

mize production processes, forecast performance, and troubleshoot problems, increasing productivity and lowering downtime [8].

2.3. Collaboration and remote access

Regardless of a manufacturing team physical location, cloud-based collaboration tools enable seamless communication and better knowledge sharing [9]. For multinational corporations with numerous locations, remote employees, or partners engaged in product development and manufacturing, this capability is especially helpful and valuable. A distributed network of people and organizations is involved in the design, production, and distribution of goods under the umbrella term "social manufacturing" [10], which describes a collaborative and decentralized manufacturing model that makes use of social networks, online communities, and open-source platforms.

2.4. Industrial Internet of Things

IIoT [11] deployments in manufacturing are greatly aided by cloud computing. Data from connected sensors and devices in factories can be sent to the cloud for processing and analysis in order to drive automation, increase equipment efficiency, and gain insights into operations.

2.5. Data Management

Manufacturers can gather, store, and analyze enormous amounts of data from a variety of sources, including sensors, equipment, supply chain systems, thanks to cloud platforms for data management and analytics [12]. Real-time monitoring, predictive maintenance, quality control, and supply chain optimization can all be done with the help of this data.

2.6. Manufacturing as a Service (MaaS)

Manufacturers can obtain on-demand production and manufacturing resources like 3D printers, CNC machining, and assembly lines thanks to cloud-based MaaS platforms [13]. Businesses can use this strategy to increase production capacity, cut costs, and quickly adapt to shifting consumer needs.

3. Background

Lately, there have been research efforts in smart manufacturing that are coupled with cloud computing, and the results have been encouraging, especially for the usage of machine/deep learning.

Huang et. al. [14] proposes a P2P cloud manufacturing method integrating deep learning and additive manufacturing to overcome long production cycles and high costs. The approach is validated by a case study on 3D printing services, which also emphasizes the need for additional advancements in printing technology, data enrichment, and algorithms optimization for more complex structures.

Although cloud computing is claimed to enable flexible resource management, IoT technologies can capture the execution dynamics. However, there is still a lack of in-depth discussion on how these technologies can be integrated to systematically handle the various dynamics of PL (Product Lifecycle) processes. In order to address the difficulties of uncertain execution dynamics in a manufacturer production and logistics process, Qu et.al. [15] suggest an IoT-based real-time production logistics synchronization system. The system enables powerful resource management and adaptive planning solutions by fusing IoT and fog manufacturing concepts. Researchers and practitioners gain important insights from the demonstration of the proposed system useful application in a large-scale paint manufacturing company using service-oriented people PL resources.

The integration of low-carbon environmental protection into Cloud Manufacturing (CMfg) systems is the main topic of the research done by Chen et.al. [16]. It takes geographic factors into account and introduces carbon emissions as a quality of service (QoS) evaluation indicator. Future research directions include utilizing historic service data, examining applications in additional optimization problems, and incorporating dynamic changes in production data and customer demand.

In the context of social manufacturing (SM), the work of Li et.al. [17] addresses the difficulty of achieving effective product customization. It suggests a powerful tool for product customization that makes use of neural radiation field techniques and large-scale pre-trained deep learning models to produce 3D content that can be printed in three dimensions. The framework also makes use of cloud computing technology to improve SM operations effectiveness by providing insightful data for the design, implementation, and optimization of cooperative production.

In the research paper by Nekovee et.al. [18], the idea of 6G-IoV (Internet of Verticals) is discussed as a transformational phase that will be facilitated by 5G infrastructure and focus on vertical industries like manufacturing, automotive, and energy. It examines the burgeoning structural and networking frameworks for 6G-IoV and presents potential futures, such as the Internet of Robotics, the Internet of Cloud Manufacturing and Smart Energy Grids. These scenarios call for significant improvements in the use of robotics, energy production, and online services in favor of renewable energy sources and sustainability.

Thames et.al. [19] has found out that opportunities in cutting-edge manufacturing and development for Industry 4.0 are driven by national initiatives in different countries. To create flexible and effective cloud-based manufacturing systems, emerging concepts like Software-Defined Cloud Manufacturing (SDCM) use IP communication technologies and abstraction. In Cloud-Based Manufacturing and Industry 4.0, SDCM aims to reduce difficulty challenges while enhancing dexterity, flexibility, and agility.

SDCM is also introduced in the study by Yang et.al. [20], to quickly reconfigure and schedule resources in a cloud manufacturing environment. SDCM enables quick method reconfiguration, operation, and evolution by transferring control logic to software. In order to effectively manage time-sensitive data traffic in SDCM, the study addresses network congestion brought on by the transmission of IoT data and suggests an optimization strategy that combines biological algorithms, Dijkstra's shortest path algorithm, and also queuing algorithms.

4. Commercial Applications

Nowadays, many technological companies offer customized solutions for Cloud Manufacturing, such as Snowflake [21], Google [22], Microsoft [23] and Amazon [24]. They all promise to improve day-to-day operations, along with substantial improvements in efficiency and quality.

The Microsoft Azure platform [23] will be the focal point of this research. The relevant framework for Smart Manufacturing, called Azure Smart Manufacturing Architecture was created to give manufacturing companies the tools they need to transform their products digitally. It has five main parts:

- **Data Ingestion and Integration**
IoT sensors, machines, and production systems are just a few of the data sources that make up this fundamental layer of data ingestion and integration. For seamless data ingestion and integration, Azure offers tools like Azure IoT Hub and Azure Data Factory.
- **Data Processing and Analytics**
Real-time data processing, analysis, and visualization is done using Azure services like Azure Stream Analytics and Azure Databricks after data has been ingested. The users can use this to monitor production processes in real time and gain actionable insights.

- **AI and Machine Learning**
The users can create predictive models for quality control, predictive maintenance, and demand forecasting using Azure's AI and machine learning capabilities, which are accessible through Azure Machine Learning. This increases operational effectiveness and lowers downtime.
- **IoT and Edge Computing**
Real-time processing and decision-making on devices and machinery is made possible by Azure IoT Edge, which extends cloud capabilities to the network's edge. For situations requiring local data processing and low-latency responses, this could prove crucial.
- **Security and Compliance**
Security is paramount in smart manufacturing. Azure offers robust security features, including Azure Security Center and Azure Sentinel, to protect sensitive manufacturing data and ensure compliance with industry regulations and standards.

A typical use case for data ingestion in a manufacturing plant, using Microsoft Azure can be seen in Figure 1.

Data is ingested using this Azure Data Factory pipeline, in order to be used with Azure Machine Learning. The user can quickly extract, transform, and load (ETL) data using Data Factory. The machine learning models can be trained in Azure Machine Learning after the data has been transformed and loaded into data storage. A dashboard, created by Microsoft PowerBI, could show the results of the ML process, providing valuable insights into the production process and forecasting.

5. Risks

Although using cloud computing for manufacturing has many advantages, there are also some risks. In this area, a few possible dangerous points will be mentioned:

- **Cybersecurity**
Malware, phishing, and hacking attempts are all common attacks on cloud-based manufacturing systems, that require firewalls and security checks.
- **Data security**
Concerns about data security and privacy arise when delicate manufacturing data is stored in the cloud. To safeguard their intellectual property and stop unauthorized access or data breaches, manufacturers must implement strong encryption, access controls, and hallmark mechanisms. Tramer [25] discusses the difficulty of protecting private machine learning (ML) models that are made available to the public through query interfaces like ML-As-A-Service systems. It emphasizes the need for better deployment strategies and countermeasures to fend off model extraction attacks, where adversaries can copy ML models without knowing their parameters or training data, potentially jeopardizing model monetization and data privacy. The author shows how effective these attacks on BigML and Amazon are, offers countermeasures, and suggest further research opportunities, evaluating these methods on production-grade ML services.
- **Dependency on Service Providers**
When using cloud services, companies are reliant on the service providers' performance, dependability, and availability, meaning that interruptions on the service could affect the manufacturing efficiency.
- **Data loss and Recovery**
For minimizing the effects of potential data loss incidents, it is essential to have reliable backup systems. These systems safeguard crucial data and operations in the event of unforeseen disruptions in addition to offering a safety net against data loss.

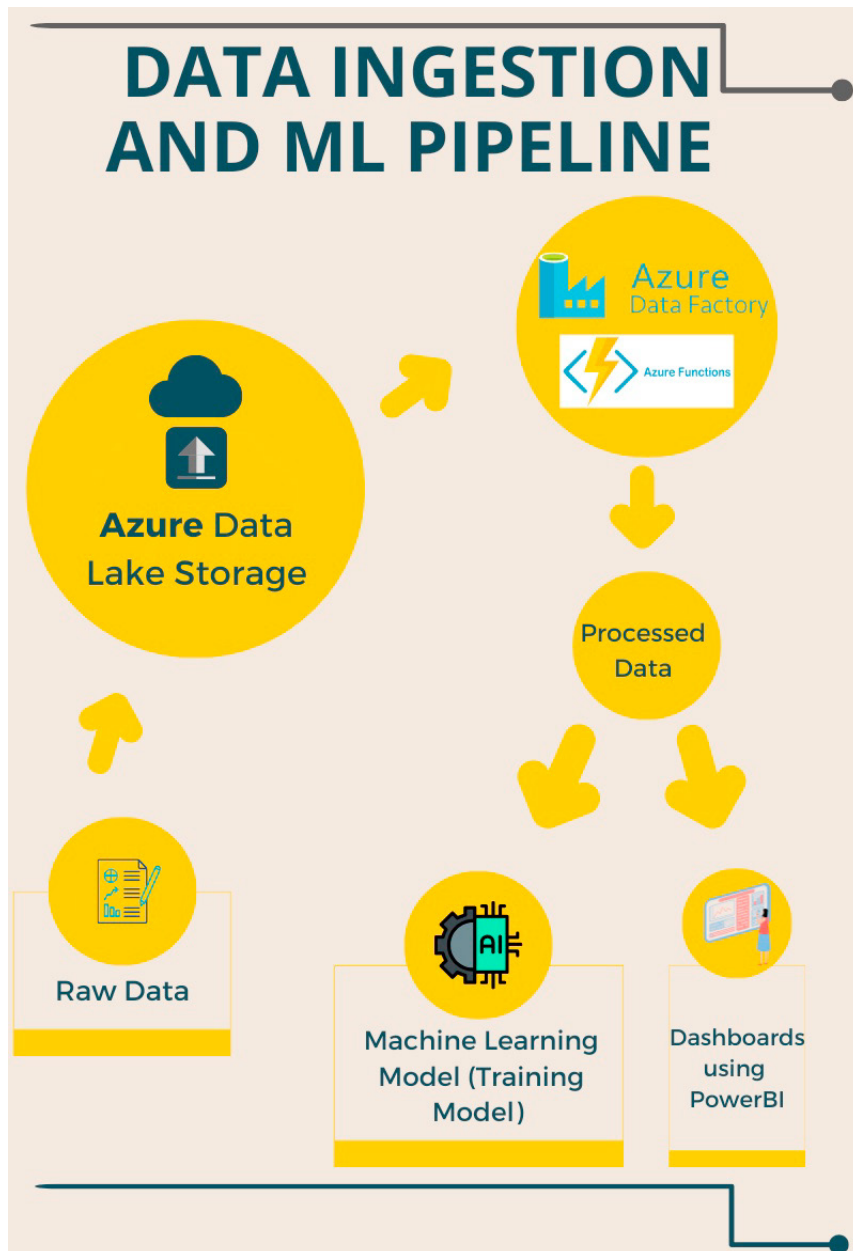


Fig. 1. Data Pipeline in Microsoft Azure

- Vendor Lock-in

Vendor Specific Platforms and protocols could hinder the switching between different providers, which frequently leads to a form of technological "lock-in." This lock-in develops as a result of each vendor's individual features and standards, which hinder compatibility and make switching to different providers or ecosystems more expensive and difficult.

6. Conclusions

To improve productivity, efficiency, and overall operational performance, the manufacturing sector is increasingly embracing cloud computing and artificial intelligence (AI). Manufacturers can use enormous amounts of data from various sources thanks to cloud computing, which offers a scalable and economical platform for data storage, processing, and analysis. Manufacturers can obtain real-time insights, optimize operations, and make data-driven decisions through cloud-based data management and analytics. Additionally, cloud-based Industrial Internet of Things (IIoT) deployments enable proactive decision-making, enable predictive maintenance, and enable manufacturers to connect and monitor their assets and machinery.

6.1. Discussions

The encouraging results of research efforts lately show that some concepts could successfully work, but the whole topic appears to be in a not very mature stage. Many companies offer commercial solutions, but there do not seem to be any completely secure solutions that have been tested in many applications, and this is expected as the whole issue is still in its early stages.

Also, there are always risks that are attached such new implementations. The data of the manufacturing plant is trusted on foreign vendors, which makes the whole operation more vulnerable to data loss, hacker attacks and vendor lock-in.

6.2. Future Research Opportunities

The capabilities of cloud computing in manufacturing are further enhanced by the new capabilities of artificial intelligence. To find patterns, foresee failures, and improve production processes, machine learning algorithms can be used on the vast data sets gathered from sensors, machines, supply chain systems. Defects and anomalies can be found using AI-powered quality control systems, which reduces waste and enhances product quality. AI is used by intelligent supply chain management systems to improve demand forecasting, streamline logistics, and optimize inventory levels.

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